Question ID 128c75e2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 128c75e2

The function g is defined by $g(x) = \frac{|x|}{a} - 14$, where $a < 0$. What is the product of $g(15a)$ and $g(7a)$?

ID: 128c75e2 Answer

Correct Answer:

609

Rationale

The correct answer is 609. It's given that the function g is defined by $gx = \frac{x}{a}$ - 14, where $a < 0$. Substituting $15a$ for x in function g yields $g15a = \frac{15a}{a}$ - 14. This function can be rewritten as $g15a = \frac{15a}{a}$ - 14, or $g15a = 15\frac{a}{a}$ - 14. Since $a < 0$, it follows that $\frac{a}{a} =$ - 1. Substituting -1 for $\frac{a}{a}$ in $g15a = 15\frac{a}{a}$ - 14 yields $g(15a) = 15($ - 1$)$ - 14, or $g(15a) =$ - 29. Similarly, substituting $7a$ for x in function g yields $g7a = \frac{7a}{a}$ - 14. This function can be rewritten as $g7a = \frac{7a}{a}$ - 14, or $g7a = 7\frac{a}{a}$ - 14. Since $a < 0$, it again follows that $\frac{a}{a} =$ - 1. Substituting -1 for $\frac{a}{a}$ in $g7a = 7\frac{a}{a}$ - 14 yields $g(7a) = 7($ - 1$)$ - 14, or $g(7a) =$ - 21. Therefore, $g(15a) =$ - 29 and $g(7a) =$ - 21. Thus, the product of $g(15a)$ and $g(7a)$ is $($ - 29$)($ - 21$)$, or 609.

Question Difficulty:

Hard

Question ID 91e7ea5e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 91e7ea5e

$h(x)=2(x-4)^2-32$

The quadratic function *h* is defined as shown. In the *xy*-plane, the graph of $y=h(x)$ intersects the *x*-axis at the points $(0,0)$ and $(t,0)$, where *t* is a constant. What is the value of *t* ?

A. 1

B. 2

C. 4

D. 8

ID: 91e7ea5e Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that the graph of $y=h(x)$ intersects the x-axis at $(0,0)$ and $(t,0)$, where t is a constant. Since this graph intersects the x-axis when $y=0$ or when $h(x)=0$, it follows that $h(0)=0$ and $h(t)=0$. If $h(t)=0$, then $0=2(t-4)^2-32$. Adding 32 to both sides of this equation yields $32=2(t-4)^2$. Dividing both sides of this equation by 2 yields $16=(t-4)^2$. Taking the square root of both sides of this equation yields $4=t-4$. Adding 4 to both sides of this equation yields $8=t$. Therefore, the value of t is 8.

Choices A, B, and C are incorrect and may result from calculation errors.

Question Difficulty:
Hard

Question ID ebed7dc6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: ebed7dc6

An auditorium has seats for $1{,}800$ people. Tickets to attend a show at the auditorium currently cost $\$4.00$. For each $\$1.00$ increase to the ticket price, 100 fewer tickets will be sold. This situation can be modeled by the equation $y = -100x^2 + 1{,}400x + 7{,}200$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. If this equation is graphed in the *xy*-plane, at what value of x is the maximum of the graph?

A. 4

B. 7

C. 14

D. 18

ID: ebed7dc6 Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that the situation can be modeled by the equation $y = -100x^2 + 1{,}400x + 7{,}200$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. Since the coefficient of the x^2 term is negative, the graph of this equation in the *xy*-plane opens downward and reaches its maximum value at its vertex. If a quadratic equation in the form $y = ax^2 + bx + c$, where a, b, and c are constants, is graphed in the *xy*-plane, the *x*-coordinate of the vertex is equal to $-\frac{b}{2a}$. For the equation $y = -100x^2 + 1{,}400x + 7{,}200$, $a = -100$, $b = 1{,}400$, and $c = 7{,}200$. It follows that the *x*-coordinate of the vertex is $-\frac{1{,}400}{2(-100)}$, or 7. Therefore, if the given equation is graphed in the *xy*-plane, the maximum of the graph occurs at an *x*-value of 7.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Hard

Question ID a9084ca4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: a9084ca4

$$f(x) = 9{,}000(0.66)^x$$

The given function f models the number of advertisements a company sent to its clients each year, where x represents the number of years since 1997, and $0 \leq x \leq 5$. If $y = f(x)$ is graphed in the *xy*-plane, which of the following is the best interpretation of the *y*-intercept of the graph in this context?

A. The minimum estimated number of advertisements the company sent to its clients during the 5 years was $1{,}708$.

B. The minimum estimated number of advertisements the company sent to its clients during the 5 years was $9{,}000$.

C. The estimated number of advertisements the company sent to its clients in 1997 was $1{,}708$.

D. The estimated number of advertisements the company sent to its clients in 1997 was $9{,}000$.

ID: a9084ca4 Answer

Correct Answer:

D

Rationale

Choice D is correct. The *y*-intercept of a graph in the *xy*-plane is the point where $x = 0$. For the given function f, the *y*-intercept of the graph of $y = fx$ in the *xy*-plane can be found by substituting 0 for x in the equation $y = 9{,}0000.66^x$, which gives $y = 9{,}0000.66^0$. This is equivalent to $y = 9{,}0001$, or $y = 9{,}000$. Therefore, the *y*-intercept of the graph of $y = fx$ is $0, 9{,}000$. It's given that the function f models the number of advertisements a company sent to its clients each year. Therefore, fx represents the estimated number of advertisements the company sent to its clients each year. It's also given that x represents the number of years since 1997. Therefore, $x = 0$ represents 0 years since 1997, or 1997. Thus, the best interpretation of the *y*-intercept of the graph of $y = fx$ is that the estimated number of advertisements the company sent to its clients in 1997 was $9{,}000$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID b8f13a3a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: b8f13a3a

Function f is defined by $f(x) = -a^x + b$, where a and b are constants. In the *xy*-plane, the graph of $y = f(x) - 12$ has a *y*-intercept at $\left(0, -\frac{75}{7}\right)$. The product of a and b is $\frac{320}{7}$. What is the value of a?

ID: b8f13a3a Answer

Correct Answer:

20

Rationale

The correct answer is 20. It's given that $fx = -a^x + b$. Substituting $-a^x + b$ for fx in the equation $y = fx - 12$ yields $y = -a^x + b - 12$. It's given that the *y*-intercept of the graph of $y = fx - 12$ is $0, -\frac{75}{7}$. Substituting 0 for x and $-\frac{75}{7}$ for y in the equation $y = -a^x + b - 12$ yields $-\frac{75}{7} = -a^0 + b - 12$, which is equivalent to $-\frac{75}{7} = -1 + b - 12$, or $-\frac{75}{7} = b - 13$. Adding 13 to both sides of this equation yields $\frac{16}{7} = b$. It's given that the product of a and b is $\frac{320}{7}$, or $ab = \frac{320}{7}$. Substituting $\frac{16}{7}$ for b in this equation yields $a\frac{16}{7} = \frac{320}{7}$. Dividing both sides of this equation by $\frac{16}{7}$ yields $a = 20$.

Question Difficulty:

Hard

Question ID 9654add7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 9654add7

$f(x) = -500x^2 + 25{,}000x$

The revenue $f(x)$, in dollars, that a company receives from sales of a product is given by the function *f* above, where *x* is the unit price, in dollars, of the product. The graph of $y = f(x)$ in the *xy*-plane intersects the *x*-axis at 0 and *a*. What does *a* represent?

A. The revenue, in dollars, when the unit price of the product is $0

B. The unit price, in dollars, of the product that will result in maximum revenue

C. The unit price, in dollars, of the product that will result in a revenue of $0

D. The maximum revenue, in dollars, that the company can make

ID: 9654add7 Answer

Correct Answer:
C

Rationale

Choice C is correct. By definition, the y-value when a function intersects the x-axis is 0. It's given that the graph of the function intersects the x-axis at 0 and a, that x is the unit price, in dollars, of a product, and that $f(x)$, where $y = f(x)$, is the revenue, in dollars, that a company receives from the sales of the product. Since the value of a occurs when $y = 0$, a is the unit price, in dollars, of the product that will result in a revenue of $0.

Choice A is incorrect. The revenue, in dollars, when the unit price of the product is $0 is represented by $f(x)$, when $x = 0$. Choice B is incorrect. The unit price, in dollars, of the product that will result in maximum revenue is represented by the x-coordinate of the maximum of f. Choice D is incorrect. The maximum revenue, in dollars, that the company can make is represented by the y-coordinate of the maximum of f.

Question Difficulty:
Hard

Question ID 263f9937

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 263f9937

Growth of a Culture of Bacteria

Day	Number of bacteria per milliliter at end of day
1	2.5×10^5
2	5.0×10^5
3	1.0×10^6

A culture of bacteria is growing at an exponential rate, as shown in the table above. At this rate, on which day would the number of bacteria per milliliter reach 5.12×10^8?

A. Day 5

B. Day 9

C. Day 11

D. Day 12

ID: 263f9937 Answer

Correct Answer:

D

Rationale

Choice D is correct. The number of bacteria per milliliter is doubling each day. For example, from day 1 to day 2, the number of bacteria increased from 2.5×10^5 to 5.0×10^5. At the end of day 3 there are 10^6 bacteria per milliliter. At the end of day 4, there will be $10^6 \times 2$ bacteria per milliliter. At the end of day 5, there will be $(10^6 \times 2) \times 2$, or $10^6 \times (2^2)$ bacteria per milliliter, and so on. At the end of day d, the number of bacteria will be $10^6 \times (2^{d-3})$. If the number of bacteria per milliliter will

reach 5.12×10^8 at the end of day d, then the equation $10^6 \times (2^{d-3}) = 5.12 \times 10^8$ must hold. Since 5.12×10^8 can be rewritten as 512×10^6, the equation is equivalent to $2^{d-3} = 512$. Rewriting 512 as 2^9 gives d – 3 = 9, so d = 12. The number of bacteria per milliliter would reach 5.12×10^8 at the end of day 12.

Choices A, B, and C are incorrect. Given the growth rate of the bacteria, the number of bacteria will not reach 5.12×10^8 per milliliter by the end of any of these days.

Question Difficulty:
Hard

Question ID 4dd4efcf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 4dd4efcf

$$f(x) = ax^2 + 4x + c$$

In the given quadratic function, a and c are constants. The graph of $y = f(x)$ in the *xy*-plane is a parabola that opens upward and has a vertex at the point (h, k), where h and k are constants. If $k < 0$ and $f(-9) = f(3)$, which of the following must be true?

I. $c < 0$

II. $a \geq 1$

A. I only

B. II only

C. I and II

D. Neither I nor II

ID: 4dd4efcf Answer

Correct Answer:

D

Rationale

Choice D is correct. It's given that the graph of $y = fx$ in the *xy*-plane is a parabola with vertex h, k. If f-9 = f3, then for the graph of $y = fx$, the point with an *x*-coordinate of -9 and the point with an *x*-coordinate of 3 have the same *y*-coordinate. In the *xy*-plane, a parabola is a symmetric graph such that when two points have the same *y*-coordinate, these points are equidistant from the vertex, and the *x*-coordinate of the vertex is halfway between the *x*-coordinates of these two points. Therefore, for the graph of $y = fx$, the points with *x*-coordinates -9 and 3 are equidistant from the vertex, h, k, and h is halfway between -9 and 3. The value that is halfway between -9 and 3 is $\frac{-9+3}{2}$, or -3. Therefore, $h =$ - 3. The equation defining f can also be written in vertex form, $fx = ax$ - $h^2 + k$. Substituting -3 for h in this equation yields $fx = ax$ - -$3^2 + k$, or $fx = ax + 3^2 + k$. This equation is equivalent to $fx = ax^2 + 6x + 9 + k$, or $fx = ax^2 + 6ax + 9a + k$. Since $fx = ax^2 + 4x + c$, it follows that $6a = 4$ and $9a + k = c$. Dividing both sides of the equation $6a = 4$ by 6 yields $a = \frac{4}{6}$, or $a = \frac{2}{3}$. Since $\frac{2}{3} < 1$, it's not true that $a \geq 1$. Therefore, statement II isn't true. Substituting $\frac{2}{3}$ for a in the equation $9a + k = c$ yields $9\frac{2}{3} + k = c$, or $6 + k = c$. Subtracting 6 from both sides of this equation yields $k = c$ - 6. If $k < 0$, then

$c - 6 < 0$, or $c < 6$. Since c could be any value less than 6, it's not necessarily true that $c < 0$. Therefore, statement I isn't necessarily true. Thus, neither I nor II must be true.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID 35e05e19

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 35e05e19

A park ranger hung squirrel houses each in the shape of a right rectangular prism for fox squirrels. Each house has a height of 11 inches. The length of each house's base is x inches, which is 1 inch more than the width of the house's base. Which function V gives the volume of each house, in cubic inches, in terms of the length of the house's base?

A. $V(x) = 11x(x - 1)$

B. $V(x) = 11x(x + 1)$

C. $V(x) = x(x + 11)(x - 1)$

D. $V(x) = x(x + 11)(x + 1)$

ID: 35e05e19 Answer

Correct Answer:

A

Rationale

Choice A is correct. The volume of a prism is equal to the area of its base times its height. It's given that the length of each house's base is x inches and that this length is 1 inch more than the width, in inches, of the house's base. It follows that the width, in inches, of the house's base is $x - 1$. The area of a rectangle is the product of its length and its width. Therefore, the area of the base of the house is $x(x - 1)$ square inches. It's given that the height of each house is 11 inches. Therefore, the function V that gives the volume of each house, in cubic inches, in terms of the length of the house's base x is $V(x) = x(x - 1)11$, or $V(x) = 11x(x - 1)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Hard

Question ID 18e35375

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 18e35375

$$f(x) = (x - 14)(x + 19)$$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

A. -266

B. -19

C. $-\frac{33}{2}$

D. $-\frac{5}{2}$

ID: 18e35375 Answer

Correct Answer:

D

Rationale

Choice D is correct. It's given that $fx = x - 14x + 19$, which can be rewritten as $fx = x^2 + 5x - 266$. Since the coefficient of the x^2-term is positive, the graph of $y = fx$ in the *xy*-plane opens upward and reaches its minimum value at its vertex. The *x*-coordinate of the vertex is the value of x such that fx reaches its minimum. For an equation in the form $fx = ax^2 + bx + c$, where a, b, and c are constants, the *x*-coordinate of the vertex is $-\frac{b}{2a}$. For the equation $fx = x^2 + 5x - 266$, $a = 1$, $b = 5$, and $c = -266$. It follows that the *x*-coordinate of the vertex is $-\frac{5}{21}$, or $-\frac{5}{2}$. Therefore, fx reaches its minimum when the value of x is $-\frac{5}{2}$.

Alternate approach: The value of x for the vertex of a parabola is the *x*-value of the midpoint between the two *x*-intercepts of the parabola. Since it's given that $fx = x - 14x + 19$, it follows that the two *x*-intercepts of the graph of $y = fx$ in the *xy*-plane occur when $x = 14$ and $x = -19$, or at the points $14, 0$ and $-19, 0$. The midpoint between two points, x_1, y_1 and x_2, y_2, is $\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}$. Therefore, the midpoint between $14, 0$ and $-19, 0$ is $\frac{14 + -19}{2}, \frac{0 + 0}{2}$, or $-\frac{5}{2}, 0$. It follows that fx reaches its minimum when the value of x is $-\frac{5}{2}$.

Choice A is incorrect. This is the *y*-coordinate of the *y*-intercept of the graph of $y = fx$ in the *xy*-plane.

Choice B is incorrect. This is one of the *x*-coordinates of the *x*-intercepts of the graph of $y = fx$ in the *xy*-plane.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID 8462b105

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 8462b105

The function f gives the product of a number, x, and a number that is 91 more than x. Which equation defines f?

A. $f(x) = x^2 + x + 91$

B. $f(x) = x^2 + 91$

C. $f(x) = x^2 + 91x$

D. $f(x) = x^2 + 91x + 91$

ID: 8462b105 Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that the function f gives the product of a number, x, and a number that is 91 more than x. A number that is 91 more than x can be represented by the expression $x + 91$. Therefore, f can be defined by the equation $f(x) = x(x + 91)$, or $f(x) = x^2 + 91x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID ce579859

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: ce579859

A model estimates that at the end of each year from 2015 to 2020, the number of squirrels in a population was 150% more than the number of squirrels in the population at the end of the previous year. The model estimates that at the end of 2016, there were 180 squirrels in the population. Which of the following equations represents this model, where n is the estimated number of squirrels in the population t years after the end of 2015 and $t \leq 5$?

A. $n = 72(1.5)^t$

B. $n = 72(2.5)^t$

C. $n = 180(1.5)^t$

D. $n = 180(2.5)^t$

ID: ce579859 Answer

Correct Answer:

B

Rationale

Choice B is correct. Since the model estimates that the number of squirrels in the population increased by a fixed percentage, 150%, each year, the model can be represented by an exponential equation of the form $n = a1 + \frac{p}{100}^t$, where a is the estimated number of squirrels in the population at the end of 2015, and the model estimates that at the end of each year, the number is $p\%$ more than the number at the end of the previous year. Since the model estimates that at the end of each year, the number was 150% more than the number at the end of the previous year, $p = 150$. Substituting 150 for p in the equation $n = a1 + \frac{p}{100}^t$ yields $n = a1 + \frac{150}{100}^t$, which is equivalent to $n = a1 + 1.5^t$, or $n = a2.5^t$. It's given that the estimated number of squirrels at the end of 2016 was 180. This means that when $t = 1$, $n = 180$. Substituting 1 for t and 180 for n in the equation $n = a2.5^t$ yields $180 = a2.5^1$, or $180 = 2.5a$. Dividing each side of this equation by 2.5 yields $72 = a$. Substituting 72 for a in the equation $n = a2.5^t$ yields $n = 722.5^t$.

Choice A is incorrect. This equation represents a model where at the end of each year, the estimated number of squirrels was 150% of, not 150% more than, the estimated number at the end of the previous year.

Choice C is incorrect. This equation represents a model where at the end of each year, the estimated number of squirrels was 150% of, not 150% more than, the estimated number at the end of the previous year, and the estimated number of squirrels at the end of 2015, not the end of 2016, was 180.

Choice D is incorrect. This equation represents a model where the estimated number of squirrels at the end of 2015, not the end of 2016, was 180.

Question Difficulty:
Hard

Question ID d139cf4b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: d139cf4b

$$f(t) = 55t - 2t^2$$

The function f is defined by the given equation. The function g is defined by $g(t) = f(t) + 3$. Which expression represents the maximum value of $g(t)$?

A. $3 + \left(\frac{55}{2}\right)^2$

B. $3 + 2\left(\frac{55}{4}\right)^2$

C. $3 - 2\left(\frac{55}{4}\right)^2$

D. $3 - \left(\frac{55}{2}\right)^2$

ID: d139cf4b Answer

Correct Answer:

B

Rationale

Choice B is correct. It's given that function g is defined by $g(t) = f(t) + 3$ and that $f(t) = 55t - 2t^2$. Substituting $55t - 2t^2$ for $f(t)$ in $g(t) = f(t) + 3$ yields $g(t) = 55t - 2t^2 + 3$, or $g(t) = -2t^2 + 55t + 3$. The maximum value of $g(t)$ can be found by completing the square to rewrite the equation defining g in the form $gt = a(t - h)^2 + k$, where the maximum value of the function is k, which occurs when $t = h$, and a is a negative constant. The equation $gt = -2t^2 + 55t + 3$ is equivalent to $g\left(t\right) = -2\left(t^2 - \frac{55}{2}t\right) + 3$, which can be rewritten as $gt = -2\left(t^2 - \frac{55}{2}t + \left(\frac{55}{4}\right)^2\right) + 3 + 2\left(\frac{55}{4}\right)^2$, or $gt = -2\left(t - \frac{55}{4}\right)^2 + 3 + 2\left(\frac{55}{4}\right)^2$. This equation is in the form $gt = a(t - h)^2 + k$, where $a = -2$, $h = \frac{55}{4}$, and $k = 3 + 2\left(\frac{55}{4}\right)^2$. Thus, the maximum value of $g(t)$ is $3 + 2\left(\frac{55}{4}\right)^2$.

Alternate approach: Since the function f is a quadratic function, the maximum value of $f(t)$ occurs at the value of t that's halfway between the two zeros of the function. The zeros of function f can be found by substituting 0 for $f(t)$ in the equation defining f, which yields $0 = 55t - 2t^2$. This equation can be rewritten as $0 = t(55 - 2t)$. By the zero product property, it follows that $t = 0$ or $55 - 2t = 0$. Subtracting 55 from each side of the equation $55 - 2t = 0$ yields $-2t = -55$. Dividing each side of this equation by -2 yields $t = \frac{55}{2}$. Therefore, the zeros of function f are 0 and $\frac{55}{2}$. The value that's halfway between 0 and $\frac{55}{2}$ can be found by calculating the average of 0 and $\frac{55}{2}$, which is $\frac{0 + \frac{55}{2}}{2}$, or $\frac{55}{4}$. It follows that the maximum of function f occurs when $t = \frac{55}{4}$. Substituting $\frac{55}{4}$ for t in the equation defining function f yields $f\left(\frac{55}{4}\right) = 55\left(\frac{55}{4}\right) - 2\left(\frac{55}{4}\right)^2$, which is equivalent to $f\left(\frac{55}{4}\right) = \frac{55^2}{4} - 2\left(\frac{55^2}{4^2}\right)$. Multiplying $\frac{55^2}{4}$ by $\frac{4}{4}$ in this equation to get a common denominator yields

$f\left(\frac{55}{4}\right) = 4\left(\frac{55^2}{4^2}\right) - 2\left(\frac{55^2}{4^2}\right)$, or $f\left(\frac{55}{4}\right) = 2\left(\frac{55^2}{4^2}\right)$, which is equivalent to $f\left(\frac{55}{4}\right) = 2\left(\frac{55}{4}\right)^2$. Thus, the maximum value of $f(t)$ is $2\left(\frac{55}{4}\right)^2$. Since the equation defining $g(t)$ is $g(t) = f(t) + 3$, the maximum value of $g(t)$ is 3 greater than the maximum value of $f(t)$. It follows that the maximum value of $g(t)$ is $3 + 2\left(\frac{55}{4}\right)^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID 9afe2370

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 9afe2370

The population P of a certain city y years after the last census is modeled by the equation below, where r is a constant and P_0 is the population when $y = 0$.

$$P = P_0(1 + r)^y$$

If during this time the population of the city decreases by a fixed percent each year, which of the following must be true?

A. $r < -1$

B. $-1 < r < 0$

C. $0 < r < 1$

D. $r > 1$

ID: 9afe2370 Answer

Correct Answer:

B

Rationale

Choice B is correct. The term (1 + r) represents a percent change. Since the population is decreasing, the percent change must be between 0% and 100%. When the percent change is expressed as a decimal rather than as a percent, the percentage change must be between 0 and 1. Because (1 + r) represents percent change, this can be expressed as $0 < 1 + r < 1$. Subtracting 1 from all three terms of this compound inequality results in $-1 < r < 0$.

Choice A is incorrect. If $r < -1$, then after 1 year, the population P would be a negative value, which is not possible. Choices C and D are incorrect. For any value of $r > 0$, $1 + r > 1$, and the exponential function models growth for positive values of the exponent. This contradicts the given information that the population is decreasing.

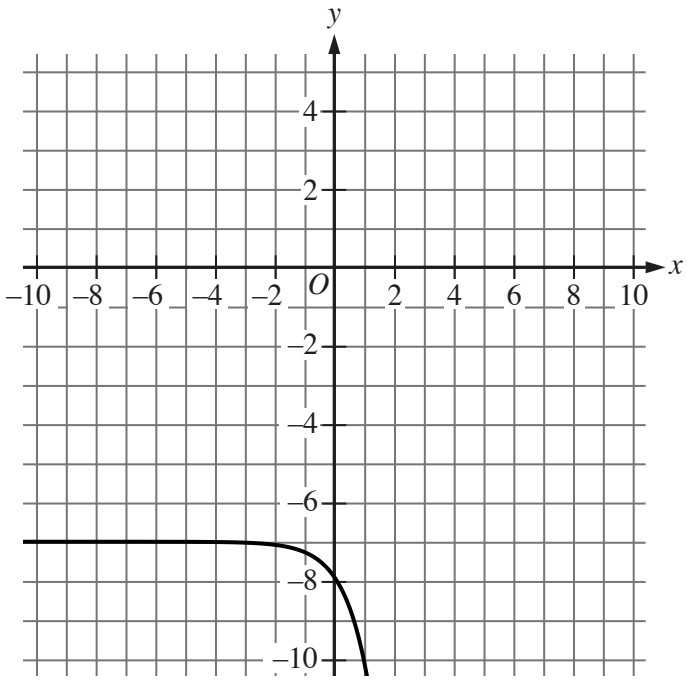
Question Difficulty:

Hard

Question ID df71424b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: df71424b



The graph of $y = f(x)$ is shown, where $f(x) = ab^x + c$, and a, b, and c are constants. For how many values of x does $f(x) = 0$?

A. Three

B. Two

C. One

D. Zero

ID: df71424b Answer

Correct Answer:

D

Rationale

Choice D is correct. Each point x, y on the graph of $y = f(x)$ in the *xy*-plane gives a value of x and its corresponding value of $f(x)$, or y. For any value of x for which $f(x) = 0$, there is a corresponding point on the graph of $y = f(x)$ with a *y*-coordinate of 0. A point with a *y*-coordinate of 0 is a point where the graph intersects the *x*-axis. It's given that $fx = ab^x + c$, where a, b, and c are constants. In the *xy*-plane, the graph of an equation of this form will lie entirely either above or below the horizontal line defined by $y = c$. The part of the graph of $y = f(x)$ shown lies entirely below the horizontal line defined by $y = -7$, and thus the entire graph of $y = f(x)$ must lie below the line defined by $y = -7$. It follows that the graph of $y = f(x)$ will never intersect the *x*-axis. Therefore, there are zero values of *x* for which $f(x) = 0$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty:
Hard

Question ID 0121a235

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 0121a235

x	$p(x)$
-2	5
-1	0
0	-3
1	-1
2	0

The table above gives selected values of a polynomial function p. Based on the values in the table, which of the following must be a factor of p ?

A. $(x-3)$

B. $(x+3)$

C. $(x-1)(x+2)$

D. $(x+1)(x-2)$

ID: 0121a235 Answer

Correct Answer:

D

Rationale

Choice D is correct. According to the table, when x is -1 or 2, $p(x)=0$. Therefore, two x-intercepts of the graph of p are $(-1,0)$ and $(2,0)$. Since $(-1,0)$ and $(2,0)$ are x-intercepts, it follows that $(x+1)$ and $(x-2)$ are factors of the polynomial equation. This is because when $x=-1$, the value of $x+1$ is 0. Similarly, when $x=2$, the value of $x-2$ is 0. Therefore, the product $(x+1)(x-2)$ is a factor of the polynomial function p.

Choice A is incorrect. The factor $x-3$ corresponds to an x-intercept of $(3,0)$, which isn't present in the table. Choice B is incorrect. The factor $x+3$ corresponds to an x-intercept of $(-3,0)$, which isn't present in the table. Choice C is incorrect. The factors $x-1$ and $x+2$ correspond to x-intercepts $(1,0)$ and $(-2,0)$, respectively, which aren't present in the table.

Question Difficulty:
Hard

Question ID bba18ecb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: bba18ecb

When the quadratic function f is graphed in the *xy*-plane, where $y = f(x)$, its vertex is $(-3, 6)$. One of the *x*-intercepts of this graph is $\left(-\frac{17}{4}, 0\right)$. What is the other *x*-intercept of the graph?

A. $\left(-\frac{29}{4}, 0\right)$

B. $\left(-\frac{7}{4}, 0\right)$

C. $\left(\frac{5}{4}, 0\right)$

D. $\left(\frac{17}{4}, 0\right)$

ID: bba18ecb Answer

Correct Answer:
B

Rationale

Choice B is correct. Since the line of symmetry for the graph of a quadratic function contains the vertex of the graph, the *x*-coordinate of the vertex of the graph of $y = fx$ is the *x*-coordinate of the midpoint of its two *x*-intercepts. The midpoint of two points with *x*-coordinates x_1 and x_2 has *x*-coordinate x_m, where $x_m = \frac{x_1 + x_2}{2}$.

It's given that the vertex is -3, 6 and one of the *x*-intercepts is $-\frac{17}{4}, 0$. Substituting -3 for x_m and $-\frac{17}{4}$ for x_1 in the equation $x_m = \frac{x_1 + x_2}{2}$ yields $-3 = \frac{-\frac{17}{4} + x_2}{2}$. Multiplying each side of this equation by 2 yields $-6 = -\frac{17}{4} + x_2$. Adding $\frac{17}{4}$ to each side of this equation yields $-\frac{7}{4} = x_2$. Therefore, the other *x*-intercept is $-\frac{7}{4}, 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID 668f1863

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 668f1863

Function f is a quadratic function where $f(-20) = 0$ and $f(-4) = 0$. The graph of $y = f(x)$ in the *xy*-plane has a vertex at $(r, -64)$. What is the value of r?

ID: 668f1863 Answer

Correct Answer:

-12

Rationale

The correct answer is -12. It's given that function f is a quadratic function where $f(-20) = 0$ and $f(-4) = 0$. It follows that the graph of $y = f(x)$ in the *xy*-plane passes through the points $(-20, 0)$ and $(-4, 0)$. When the graph of a quadratic function contains two points $(a, 0)$ and $(b, 0)$, the *x*-coordinate of the vertex of the graph is the average of a and b. Therefore, the *x*-coordinate of the vertex of the graph of $y = f(x)$ is $\frac{-20 + -4}{2}$, or -12. It's given that the graph of $y = f(x)$ in the *xy*-plane has a vertex at $(r, -64)$. It follows that the value of r is -12.

Question Difficulty:

Hard

Question ID 70753f99

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 70753f99

The function f is defined by $f(x)=(x+3)(x+1)$. The graph of f in the xy-plane is a parabola. Which of the following intervals contains the x-coordinate of the vertex of the graph of f?

A. $-4<x<-3$

B. $-3<x<1$

C. $1<x<3$

D. $3<x<4$

ID: 70753f99 Answer

Correct Answer:

B

Rationale

Choice B is correct. The graph of a quadratic function in the xy-plane is a parabola. The axis of symmetry of the parabola passes through the vertex of the parabola. Therefore, the vertex of the parabola and the midpoint of the segment between the two x-intercepts of the graph have the same x-coordinate. Since $f(-3)=f(-1)=0$, the x-coordinate of the vertex is $\frac{(-3)+(-1)}{2}=-2$. Of the shown intervals, only the interval in choice B contains -2. Choices A, C, and D are incorrect and may result from either calculation errors or misidentification of the graph's x-intercepts.

Question Difficulty:

Hard

Question ID 58dcc59f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 58dcc59f

A landscaper is designing a rectangular garden. The length of the garden is to be 5 feet longer than the width. If the area of the garden will be 104 square feet, what will be the length, in feet, of the garden?

ID: 58dcc59f Answer

Rationale

The correct answer is 13. Let w represent the width of the rectangular garden, in feet. Since the length of the garden will be 5 feet longer than the width of the garden, the length of the garden will be $w+5$ feet. Thus the area of the garden will be $w(w+5)$. It is also given that the area of the garden will be 104 square feet. Therefore, $w(w+5)=104$, which is equivalent to $w^2+5w-104=0$. Factoring this equation results in $(w+13)(w-8)=0$. Therefore, $w=8$ and $w=-13$. Because width cannot be negative, the width of the garden must be 8 feet. This means the length of the garden must be $8+5=13$ feet.

Question Difficulty:
Hard

Question ID 84dd43f8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 84dd43f8

For the function f, $f(0) = 86$, and for each increase in x by 1, the value of $f(x)$ decreases by 80%. What is the value of $f(2)$?

ID: 84dd43f8 Answer

Correct Answer:
3.44, 86/25

Rationale

The correct answer is 3.44. It's given that $f0 = 86$ and that for each increase in x by 1, the value of fx decreases by 80%. Because the output of the function decreases by a constant percentage for each 1-unit increase in the value of x, this relationship can be represented by an exponential function of the form $fx = ab^x$, where a represents the initial value of the function and b represents the rate of decay,
expressed as a decimal. Because $f0 = 86$, the value of a must be 86. Because the value of fx decreases by 80% for each 1-unit increase in x, the value of b must be $(1 - 0.80)$, or 0.2. Therefore, the function f can be defined by $fx = 860.2^x$. Substituting 2 for x in this function yields $f2 = 860.2^2$, which is equivalent to $f2 = 860.04$, or $f2 = 3.44$. Either 3.44 or $86 / 25$ may be entered as the correct answer.

Alternate approach: It's given that $f0 = 86$ and that for each increase in x by 1, the value of fx decreases by 80%. Therefore, when $x = 1$, the value of fx is $(100 - 80)\%$, or 20%, of 86, which can be expressed as 0.2086. Since $0.2086 = 17.2$, the value of $f1$ is 17.2. Similarly, when $x = 2$, the value of fx is 20% of 17.2, which can be expressed as $0.2017.2$. Since $0.2017.2 = 3.44$, the value of $f2$ is 3.44. Either 3.44 or $86 / 25$ may be entered as the correct answer.

Question Difficulty:
Hard

Question ID 59d1f4b5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 59d1f4b5

$M = 1{,}800(1.02)^t$

The equation above models the number of members, *M*, of a gym *t* years after the gym opens. Of the following, which equation models the number of members of the gym *q* quarter years after the gym opens?

A. $M = 1{,}800(1.02)^{\frac{q}{4}}$

B. $M = 1{,}800(1.02)^{4q}$

C. $M = 1{,}800(1.005)^{4q}$

D. $M = 1{,}800(1.082)^{q}$

ID: 59d1f4b5 Answer

Correct Answer:

A

Rationale

Choice A is correct. In 1 year, there are 4 quarter years, so the number of quarter years, q, is 4 times the number of years, t ; that is, $q = 4t$. This is equivalent to $t = \frac{q}{4}$, and substituting this into the expression for M in terms of t gives $M = 1{,}800(1.02)^{\frac{q}{4}}$.

Choices B and D are incorrect and may be the result of incorrectly using $t = 4q$ instead of $q = 4t$. (Choices B and D are nearly the same since 1.02^{4q} is equivalent to $(1.02^4)^q$, which is approximately 1.082^q.) Choice C is incorrect and may be the result of incorrectly using $t = 4q$ and unnecessarily dividing 0.02 by 4.

Question Difficulty:
Hard

Question ID 635f54ee

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 635f54ee

The surface area of a cube is $6\left(\frac{a}{4}\right)^2$, where a is a positive constant. Which of the following gives the perimeter of one face of the cube?

A. $\frac{a}{4}$

B. a

C. $4a$

D. $6a$

ID: 635f54ee Answer

Correct Answer:

B

Rationale

Choice B is correct. A cube has 6 faces of equal area, so if the total surface area of a cube is $6\left(\frac{a}{4}\right)^2$, then the area of one face is $\left(\frac{a}{4}\right)^2$. Likewise, the area of one face of a cube is the square of one of its edges; therefore, if the area of one face is $\left(\frac{a}{4}\right)^2$, then the length of one edge of the cube is $\frac{a}{4}$. Since the perimeter of one face of a cube is four times the length of one edge, the perimeter is $4\left(\frac{a}{4}\right)=a$.

Choice A is incorrect because if the perimeter of one face of the cube is $\frac{a}{4}$, then the total surface area of the cube is $6\left(\frac{\frac{a}{4}}{4}\right)^2 = 6\left(\frac{a}{16}\right)^2$, which is not $6\left(\frac{a}{4}\right)^2$.

Choice C is incorrect because if the perimeter of one face of the cube is 4a, then the total surface area of the cube is $6\left(\frac{4a}{4}\right)^2 = 6a^2$, which is not $6\left(\frac{a}{4}\right)^2$. Choice D is incorrect because if the perimeter of one face of the cube is 6a, then the total surface area of the cube is $6\left(\frac{6a}{4}\right)^2 = 6\left(\frac{3a}{2}\right)^2$, which is not $6\left(\frac{a}{4}\right)^2$.

Question Difficulty:
Hard

Question ID de39858a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: de39858a

The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy-plane passes through the points $(0, 10)$ and $(-2, \frac{325}{36})$. What is the value of ab?

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. 54

D. 60

ID: de39858a Answer

Correct Answer:

C

Rationale

Choice C is correct. It's given that the function h is defined by $hx = a^x + b$ and that the graph of $y = h(x)$ in the *xy*-plane passes through the points $0, 10$ and -2, $\frac{325}{36}$. Substituting 0 for x and 10 for hx in the equation $hx = a^x + b$ yields $10 = a^0 + b$, or $10 = 1 + b$. Subtracting 1 from both sides of this equation yields $9 = b$. Substituting -2 for x and $\frac{325}{36}$ for hx in the equation $h(x) = a^x + 9$ yields $\frac{325}{36} = a^{-2} + 9$. Subtracting 9 from both sides of this equation yields $\frac{1}{36} = a^{-2}$, which can be rewritten as $a^2 = 36$. Taking the square root of both sides of this equation yields $a = 6$ and $a = -6$, but because it's given that a is a positive constant, a must equal 6. Because the value of a is 6 and the value of b is 9, the value of ab is $(6)(9)$, or 54.

Choice A is incorrect and may result from finding the value of $a^{-2}b$ rather than the value of ab.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from correctly finding the value of a as 6, but multiplying it by the *y*-value in the first ordered pair rather than by the value of b.

Question Difficulty:

Hard

Question ID 1178f2df

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 1178f2df

x	y
21	-8
23	8
25	-8

The table shows three values of x and their corresponding values of y, where $y = f(x) + 4$ and f is a quadratic function. What is the *y*-coordinate of the *y*-intercept of the graph of $y = f(x)$ in the *xy*-plane?

ID: 1178f2df Answer

Correct Answer:

-2112

Rationale

The correct answer is -2,112. It's given that f is a quadratic function. It follows that f can be defined by an equation of the form $fx = ax - h^2 + k$, where a, h, and k are constants. It's also given that the table shows three values of x and their corresponding values of y, where $y = fx + 4$. Substituting $ax - h^2 + k$ for fx in this equation yields $y = ax - h^2 + k + 4$. This equation represents a quadratic relationship between x and y, where $k + 4$ is either the maximum or the minimum value of y, which occurs when $x = h$. For quadratic relationships between x and y, the maximum or minimum value of y occurs at the value of x halfway between any two values of x that have the same corresponding value of y. The table shows that *x*-values of 21 and 25 correspond to the same *y*-value, -8. Since 23 is halfway between 21 and 25, the maximum or minimum value of y occurs at an *x*-value of 23. The table shows that when $x = 23$, $y = 8$. It follows that $h = 23$ and $k + 4 = 8$. Subtracting 4 from both sides of the equation $k + 4 = 8$ yields $k = 4$. Substituting 23 for h and 4 for k in the equation $y = ax - h^2 + k + 4$ yields $y = ax - 23^2 + 4 + 4$, or $y = ax - 23^2 + 8$. The value of a can be found by substituting any *x*-value and its corresponding *y*-value for x and y, respectively, in this equation. Substituting 25 for x and -8 for y in this equation yields $-8 = a25 - 23^2 + 8$, or $-8 = a2^2 + 8$. Subtracting 8 from both sides of this equation yields $-16 = a2^2$, or $-16 = 4a$. Dividing both sides of this equation by 4 yields $-4 = a$. Substituting -4 for a, 23 for h, and 4 for k in the equation $fx = ax - h^2 + k$ yields $fx = -4x - 23^2 + 4$. The *y*-intercept of the graph of $y = fx$ in the *xy*-plane is the point on the graph where $x = 0$. Substituting 0 for x in the equation $fx = -4x - 23^2 + 4$ yields $f0 = -40 - 23^2 + 4$, or $f0 = -4-23^2 + 4$. This is equivalent to $f0 = -2{,}112$, so the *y*-intercept of the graph of $y = fx$ in the *xy*-plane is $0, -2{,}112$. Thus, the *y*-coordinate of the *y*-intercept of the graph of $y = fx$ in the *xy*-plane is -2,112.

Question Difficulty:
Hard

Question ID 84e8cc72

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 84e8cc72

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object is launched off an elevated surface. The model indicates the object has an initial height of 10 feet above the ground and reaches its maximum height of $1{,}034$ feet above the ground 8 seconds after being launched. Based on the model, what is the height, in feet, of the object above the ground 10 seconds after being launched?

A. 234

B. 778

C. 970

D. $1{,}014$

ID: 84e8cc72 Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that a quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object is launched off an elevated surface. This quadratic function can be defined by an equation of the form $fx = ax - h^2 + k$, where fx is the height of the object x seconds after it was launched, and a, h, and k are constants such that the function reaches its maximum value, k, when $x = h$. Since the model indicates the object reaches its maximum height of $1{,}034$ feet above the ground 8 seconds after being launched, fx reaches its maximum value, $1{,}034$, when $x = 8$. Therefore, $k = 1{,}034$ and $h = 8$. Substituting 8 for h and $1{,}034$ for k in the function $fx = ax - h^2 + k$ yields $fx = ax - 8^2 + 1{,}034$. Since the model indicates the object has an initial height of 10 feet above the ground, the value of fx is 10 when $x = 0$. Substituting 0 for x and 10 for fx in the equation $fx = ax - 8^2 + 1{,}034$ yields $10 = a0 - 8^2 + 1{,}034$, or $10 = 64a + 1{,}034$. Subtracting $1{,}034$ from both sides of this equation yields $64a = -1{,}024$. Dividing both sides of this equation by 64 yields $a = -16$. Therefore, the model can be represented by the equation $fx = -16x - 8^2 + 1{,}034$. Substituting 10 for x in this equation yields $f10 = -1610 - 8^2 + 1{,}034$, or $f10 = 970$. Therefore, based on the model, 10 seconds after being launched, the height of the object above the ground is 970 feet.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID 6d9e01a2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 6d9e01a2

$$f(x) = 4x^2 - 50x + 126$$

The given equation defines the function f. For what value of x does $f(x)$ reach its minimum?

ID: 6d9e01a2 Answer

Correct Answer:

25/4, 6.25

Rationale

The correct answer is $\frac{25}{4}$. The given equation can be rewritten in the form $fx = ax - h^2 + k$, where a, h, and k are constants. When $a > 0$, h is the value of x for which fx reaches its minimum. The given equation can be rewritten as $fx = 4x^2 - \frac{50}{4}x + 126$, which is equivalent to $fx = 4x^2 - \frac{50}{4}x + \frac{50}{8}^2 - \frac{50}{8}^2 + 126$. This equation can be rewritten as $fx = 4x - \frac{50}{8}^2 - \frac{50}{8}^2 + 126$, or $fx = 4x - \frac{50}{8}^2 - 4\frac{50}{8}^2 + 126$, which is equivalent to $fx = 4x - \frac{25}{4}^2 - \frac{121}{4}$. Therefore, $h = \frac{25}{4}$, so the value of x for which fx reaches its minimum is $\frac{25}{4}$. Note that 25/4 and 6.25 are examples of ways to enter a correct answer.

Question Difficulty:

Hard

Question ID 9f2ecade

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 9f2ecade

$$h(x) = x^3 + ax^2 + bx + c$$

The function h is defined above, where a, b, and c are integer constants. If the zeros of the function are -5, 6, and 7, what is the value of c ?

ID: 9f2ecade Answer

Rationale

The correct answer is 210. Since -5, 6, and 7 are zeros of the function, the function can be rewritten as $h(x) = (x+5)(x-6)(x-7)$. Expanding the function yields $h(x) = x^3 - 8x^2 - 23x + 210$. Thus, $a = -8$, $b = -23$, and $c = 210$. Therefore, the value of c is 210.

Question Difficulty:
Hard

Question ID 1fe10d97

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 1fe10d97

$$p(t) = 90{,}000(1.06)^t$$

The given function p models the population of Lowell t years after a census. Which of the following functions best models the population of Lowell m months after the census?

A. $r(m) = \frac{90{,}000}{12}(1.06)^m$

B. $r(m) = 90{,}000\left(\frac{1.06}{12}\right)^m$

C. $r(m) = 90{,}000\left(\frac{1.06}{12}\right)^{\frac{m}{12}}$

D. $r(m) = 90{,}000(1.06)^{\frac{m}{12}}$

ID: 1fe10d97 Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that the function p models the population of Lowell t years after a census. Since there are 12 months in a year, m months after the census is equivalent to $\frac{m}{12}$ years after the census. Substituting $\frac{m}{12}$ for t in the equation $pt = 90{,}0001.06^t$ yields $p\frac{m}{12} = 90{,}0001.06^{\frac{m}{12}}$. Therefore, the function r that best models the population of Lowell m months after the census is $rm = 90{,}0001.06^{\frac{m}{12}}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID b73ee6cf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: b73ee6cf

The population of a town is currently 50,000, and the population is estimated to increase each year by 3% from the previous year. Which of the following equations can be used to estimate the number of years, *t*, it will take for the population of the town to reach 60,000 ?

A. $50{,}000 = 60{,}000(0.03)^t$

B. $50{,}000 = 60{,}000(3)^t$

C. $60{,}000 = 50{,}000(0.03)^t$

D. $60{,}000 = 50{,}000(1.03)^t$

ID: b73ee6cf Answer

Correct Answer:
D

Rationale

Choice D is correct. Stating that the population will increase each year by 3% from the previous year is equivalent to saying that the population each year will be 103% of the population the year before. Since the initial population is 50,000, the population after t years is given by $50{,}000(1.03)^t$. It follows that the equation $60{,}000 = 50{,}000(1.03)^t$ can be used to estimate the number of years it will take for the population to reach 60,000.

Choice A is incorrect. This equation models how long it will take the population to decrease from 60,000 to 50,000, which is impossible given the growth factor. Choice B is incorrect and may result from misinterpreting a 3% growth as growth by a factor of 3. Additionally, this equation attempts to model how long it will take the population to decrease from 60,000 to 50,000. Choice C is incorrect and may result from misunderstanding how to model percent growth by multiplying the initial amount by a factor greater than 1.

Question Difficulty:
Hard

Question ID 7eed640d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 7eed640d

$h(x) = -16x^2 + 100x + 10$

The quadratic function above models the height above the ground *h*, in feet, of a projectile *x* seconds after it had been launched vertically. If $y = h(x)$ is graphed in the *xy*-plane, which of the following represents the real-life meaning of the positive *x*-intercept of the graph?

A. The initial height of the projectile

B. The maximum height of the projectile

C. The time at which the projectile reaches its maximum height

D. The time at which the projectile hits the ground

ID: 7eed640d Answer

Correct Answer:
D

Rationale

Choice D is correct. The positive x-intercept of the graph of $y = h(x)$ is a point (x,y) for which $y = 0$. Since $y = h(x)$ models the height above the ground, in feet, of the projectile, a y-value of 0 must correspond to the height of the projectile when it is 0 feet above ground or, in other words, when the projectile is on the ground. Since x represents the time since the projectile was launched, it follows that the positive x-intercept, $(x,0)$, represents the time at which the projectile hits the ground.

Choice A is incorrect and may result from misidentifying the y-intercept as a positive x-intercept. Choice B is incorrect and may result from misidentifying the y-value of the vertex of the graph of the function as an x-intercept. Choice C is incorrect and may result from misidentifying the x-value of the vertex of the graph of the function as an x-intercept.

Question Difficulty:
Hard

Question ID 43926bd9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 43926bd9

x	$f(x)$
1	a
2	a^5
3	a^9

For the exponential function *f*, the table above shows several values of *x* and their corresponding values of $f(x)$, where *a* is a constant greater than 1. If *k* is a constant and $f(k) = a^{29}$, what is the value of *k* ?

ID: 43926bd9 Answer

Rationale

The correct answer is 8. The values of $f(x)$ for the exponential function f shown in the table increase by a factor of a^4 for each increase of 1 in x. This relationship can be represented by the equation $f(x) = a^{4x+b}$, where b is a constant. It's given that when $x = 2$, $f(x) = a^5$. Substituting 2 for x and a^5 for $f(x)$ into $f(x) = a^{4x+b}$ yields $a^5 = a^{4(2)+b}$. Since $4(2)+b = 5$, it follows that $b = -3$. Thus, an equation that defines the function f is $f(x) = a^{4x-3}$. It follows that the value of k such that $f(k) = a^{29}$ can be found by solving the equation $4k-3 = 29$, which yields $k = 8$.

Question Difficulty:
Hard

Question ID f25a34aa

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: f25a34aa

The area of a triangle is equal to x^2 square centimeters. The length of the base of the triangle is $2x + 22$ centimeters, and the height of the triangle is $x - 10$ centimeters. What is the value of x?

ID: f25a34aa Answer

Correct Answer:
110

Rationale

The correct answer is 110. The area of a triangle is equal to one half of the product of the length of the base of the triangle and the height of the triangle. It's given that the length of the base of the triangle is $2x + 22$ centimeters and the height of the triangle is $x - 10$ centimeters; therefore, its area is $\frac{1}{2}\left(2x + 22\right)\left(x - 10\right)$ square centimeters. It's also given that the area of the triangle is equal to x^2 square centimeters. Therefore, it follows that $\frac{1}{2}\left(2x + 22\right)\left(x - 10\right) = x^2$. This equation can be rewritten as $\left(x + 11\right)\left(x - 10\right) = x^2$, or $x^2 + x - 110 = x^2$. Subtracting x^2 from both sides of this equation yields $x - 110 = 0$. Adding 110 to both sides of this equation yields $x = 110$. Therefore, the value of x is 110.

Question Difficulty:
Hard

Question ID a58232b7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: a58232b7

The functions g and h are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

I. $g(x) = 18(1.16)(1.4)^{x+2}$

II. $h(x) = 18(1.4)^{x+4}$

A. I only

B. II only

C. I and II

D. Neither I nor II

ID: a58232b7 Answer

Correct Answer:

D

Rationale

Choice D is correct. A function defined by an equation in the form $fx = ab^{x+h}$, where a, b, and h are positive constants and $x \geq 0$, has a minimum value of $f(0)$. It's given that function g is defined by $gx = 181.161.4^{x+2}$, which is equivalent to $gx = 20.881.4^{x+2}$. Substituting 0 for x in this equation yields $g0 = 20.88(1.4)^{0+2}$, or $g(0) = 40.9248$. Therefore, the minimum value of $g(x)$ is 40.9248, so $gx = 181.161.4^{x+2}$ doesn't display its minimum value as a constant or coefficient. It's also given that function h is defined by $hx = 181.4^{x+4}$. Substituting 0 for x in this equation yields $h0 = 181.4^{0+4}$, or $h(0) = 69.1488$. Therefore, the minimum value of $h(x)$ is 69.1488, so $hx = 181.4^{x+4}$ doesn't display its minimum value as a constant or coefficient. Therefore, neither I nor II displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID a7711fe8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: a7711fe8

What is the minimum value of the function f defined by $f(x)=(x-2)^2-4$?

A. -4

B. -2

C. 2

D. 4

ID: a7711fe8 Answer

Correct Answer:

A

Rationale

Choice A is correct. The given quadratic function f is in vertex form, $f(x)=(x-h)^2+k$, where (h,k) is the vertex of the graph of $y=f(x)$ in the xy-plane. Therefore, the vertex of the graph of $y=f(x)$ is $(2,-4)$. In addition, the y-coordinate of the vertex represents either the minimum or maximum value of a quadratic function, depending on whether the graph of the function opens upward or downward. Since the leading coefficient of f (the coefficient of the term $(x-2)^2$) is 1, which is positive, the graph of $y=f(x)$ opens upward. It follows that at $x=2$, the minimum value of the function f is -4.

Choice B is incorrect and may result from making a sign error and from using the x-coordinate of the vertex. Choice C is incorrect and may result from using the x-coordinate of the vertex. Choice D is incorrect and may result from making a sign error.

Question Difficulty:

Hard

Question ID 1a722d7d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 1a722d7d

Let the function p be defined as $p(x)=\frac{(x-c)^2+160}{2c}$, where c is a constant. If $p(c)=10$, what is the value of $p(12)$?

A. 10.00

B. 10.25

C. 10.75

D. 11.00

ID: 1a722d7d Answer

Correct Answer:

D

Rationale

Choice D is correct. The value of p(12) depends on the value of the constant c, so the value of c must first be determined. It is given that p(c) = 10. Based on the definition of p, it follows that:

$$p(c)=\frac{(c-c)^2+160}{2c}=10$$

$$\frac{160}{2c}=10$$

$$2c=16$$

$c = 8$

This means that $p(x) = \frac{(x-8)^2+160}{16}$ for all values of x. Therefore:

$$p(12) = \frac{(12-8)^2+160}{16}$$

$$= \frac{16+160}{16}$$

$$= 11$$

Choice A is incorrect. It is the value of p(8), not p(12). Choices B and C are incorrect. If one of these values were correct, then x = 12 and the selected value of p(12) could be substituted into the equation to solve for c. However, the values of c that result from choices B and C each result in p(c) < 10.

Question Difficulty:
Hard